Saksham Hans

Software Engineer | Full-Stack & AI/ML

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Education

ABV-IIITM Gwalior – B.Tech. in Computer Science and Engineering July 2024 – Expected 2028
CGPA: 9.34/10 | Semester III GPA: 10.00/10

Relevant Coursework: Data Structures, Design and Analysis of Algorithms, Object-Oriented Programming, Operating Systems, Computer Networks, Databases, Software Engineering, Compiler Design, Probability and Statistics, Data Analytics and Visualization, Trustworthy AI

Skills Summary

- **Languages:** C++, Java, Python, JavaScript, TypeScript, SQL, C
- **Backend:** Node.js, Express.js, FastAPI, Spring Boot, REST APIs, WebSockets
- **Cloud / DevOps:** AWS, Docker, GitHub Actions (CI/CD), Git, GitHub Apps, Webhooks, Postman
- **Databases:** PostgreSQL, MySQL, MongoDB, SQLite
- **ML / Data:** scikit-learn, Sentence Transformers, NumPy, Pandas, CVXPY
- **Frontend:** React, Next.js, Tailwind CSS, HTML5, CSS3

Projects

DeployGuard | *Node.js, Python, FastAPI, React, PostgreSQL, scikit-learn, GitHub Actions* GitHub | Live Demo

- Built a GitHub App that flags bundle-size and query regressions, posting Check Runs and PR comments with chunk-level diffs.
- Trained a MiniLM + Logistic Regression classifier across 10 root-cause classes, achieving 0.82 ± 0.04 macro-F1; integrated a Groq LLM fallback with 14 ms mean / 16 ms P95 local inference.
- Added idempotent webhooks with dedup and backoff retries; fixed a bug flagging a 33.7% bundle-size cut as a regression.

ZeroPack | *Node.js, JavaScript (Zero-Dependency 72-Hour Hackathon)* GitHub | Live Demo

- Developed a zero-dependency JavaScript bundler, minifier, and RFC 6455 WebSocket HMR server using only Node.js core APIs.
- Implemented an incremental dependency-graph cache for 500 ESM modules, achieving 34.38 ms rebuilds and a 3.6× speedup.
- Created a state-machine minifier reaching 52.55 MB/s; 18 test suites and 100% reproducible builds (SHA-256 verified).

Contrib Compass 2.0 | *Next.js, Express, Groq, Sentence Transformers, SQLite* GitHub | Live Demo

- Rebuilt an open-source issue triage and contributor-matching platform, using Groq LLM classification and 768-D sentence embeddings to route issues to suitable contributors.
- Reached 94.3% triage agreement (33/35 real issues), +28.6 percentage points over heuristic baselines, at ~312 ms average latency.
- Built a maintainer feedback loop that stores corrections and recalibrates future issue classifications.

Experience

InterviewHire | *Software Engineering Intern* June 2026 – Present

- Contributed to automated CI/CD deployment pipelines using AWS and GitHub Actions, containerizing local and staging environments with Docker to standardize developer workflows and ensure consistent, reproducible builds across deployments.
- Implemented and maintained 5–8+ modular backend routes following SOLID principles, refactoring duplicated logic to improve maintainability, and built 10+ reusable React components with lazy loading to reduce overall frontend bundle size by 15–25%.
- Fixed 3+ security vulnerabilities (e.g., exposed environment secrets and missing input validation) during pre-production deployment hardening and collaborated within an 8-member Agile engineering team, validating production releases through peer code reviews and automated test suites.

Apicurio Registry | *Open-Source Contributor (MCP module, Java)* Pull Requests

- Authored 4 pull requests (2 merged, 1 approved, 1 in maintainer review) hardening how the MCP server handles untrusted input from AI agents; added VersionState filtering so withdrawn and draft prompt template versions no longer appear in agent-facing discovery.
- Replaced exception-throwing enum parsing with case-insensitive validation across 7 tool and search paths, with regression tests; incorporated maintainer review feedback, including a call site missed in the first revision.

Positions of Responsibility

Vice Secretary | *IEEE Student Branch, ABV-IIITM Gwalior* January 2026 – Present

- Coordinated workshops, hackathons, conferences, and technical events as Vice Secretary of the IEEE Student Branch.
- **IATMSI 2026:** Gathered feedback from 100+ delegates across events to improve logistics and participant experience for future events.
- **Hacksagon 2026:** Designed and executed accommodation logistics for 900+ participants from across India.

Achievements & Certifications

- **Top 0.3%:** Amazon ML Challenge 2026, ranking Top 250 among 90,000+ participants.
- **Finalist:** Top 30 Teams, Morrow 1.0 Hackathon, selected from 1,500+ participants for the final round. Certificate
- **Top 10%:** Zero-Dependency 72-Hour Hackathon, among 4,200+ registrants. Certificate
- **Competitive Programming:** Reached a 1700+ LeetCode rating with 450+ DSA problems solved.
- **Mathematics Olympiad:** Ranked 7th internationally in the SOF International Mathematics Olympiad (IMO).
- BCG Data Science Job Simulation: Performed customer churn analysis using Python (Pandas, NumPy), built a Random Forest model achieving 50% recall, visualized key trends, and presented actionable business insights. Certificate
- JPMorgan Chase Software Engineering Job Simulation: Integrated Kafka with Spring Boot for event-driven transaction processing, with JPA validation and REST APIs. Certificate
- Goldman Sachs Job Simulations: Completed Operations, Risk, Internal Audit, and Controllers simulations covering financial analysis, risk assessment, trade settlement, NAV valuation, reconciliation, and Excel-based reporting. Certificate